Mubashara Rehman

AI Researcher • Computer Vision & Medical Imaging

PhD Candidate, Awaiting Final Defence (Thesis Passed External Review)

University of Udine, Italy | IRCCS Centro di Riferimento Oncologico di Aviano, Italy

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Website | **LinkedIn** | **Google Scholar** | **ORCID** | **GitHub** | **Timeline**

Professional Profile

PhD candidate in Artificial Intelligence (thesis passed external review; final defence pending) with expertise in computer vision, deep learning and medical image analysis. Experienced in designing and developing generative AI models for CT and MRI reconstruction, image enhancement and cross-modality domain translation using PyTorch. Proven ability to collaborate across academia, healthcare and industry (including IRCCS CRO Aviano (Italy), Leiden University Medical Center (LUMC), and Philips Research) to deliver innovative AI solutions for medical imaging. Looking ahead, I aim to further advance my research in AI while exploring emerging areas such as Neuro-AI, multimodal learning, foundation models, and intelligent systems, with a focus on developing AI that can address complex real-world challenges.

Research Interests

- Deep Learning for Medical Imaging
- Metal Artifact Reduction in CT
- CT Domain Transformation (kVCT → MVCT)
- AI-driven CT Reconstruction & MRI Enhancement

Education

Ph.D. in Artificial Intelligence and Data Analysis for Personalised Oncology — Sep 2023 – Present

Università degli Studi di Udine | Udine, Italy

Awaiting final defence • External review passed 29 Oct 2026 • Thesis submitted 31 Aug 2026

Dept. of Mathematics, Computer Science and Physics (DMIF), in collaboration with IRCCS CRO Aviano

Advisor: Prof. Christian Micheloni • Co-Advisor: Prof. Niki Martinel (Machine Learning and Perception Lab)

Collaborators: Dr. Michele Avanzo, Dr. Riccardo Spizzo (IRCCS CRO Aviano)

- Research on deep learning methods for metal artifact reduction in CT images for radiotherapy applications.
- Developed kVCT-to-MVCT domain transformation frameworks to improve image quality and support radiotherapy treatment planning.

M.Sc. in Software Engineering — Sep 2017 – Nov 2021

Tianjin University, College of Intelligence and Computing | Tianjin, China

- Graduated with 83.2%; awarded the Tianjin University International Student Scholarship.

B.Sc. in Software Engineering — Sep 2012 – Jun 2016

Fatima Jinnah Women University | Rawalpindi, Pakistan

- Final grade: 82%; degree verified by World Education Services (WES).
- Senior design thesis: *"Design and Development of an Android Application for Indoor Navigation and Positioning System using Sensor Networks"* – graded A+.

Higher Secondary School Certificate (HSSC) – Pre-Engineering — 2010 – 2012

Grades 11 and 12 | Grade A

Secondary School Certificate (SSC) – Science — 2008 – 2010

Grades 9 and 10 | Grade A

Research Experience

PhD Visiting Researcher — Jan 2026 – Jun 2026

Leiden University Medical Center (LUMC) | Leiden, Netherlands

Supervisor: Prof. Marius Staring • In collaboration with Dr. Mariya Doneva (Philips Research) and Dr. Yiming Dong (LUMC)

Partners: Division of Image Processing (LKEB) and C.J. Gorter Center for High Field MRI, LUMC; Philips Research, Netherlands

- Researched deep learning-based low-field to high-field prostate MRI enhancement in a joint LUMC–Philips Research project.
- Developed novel generative AI methods for MRI reconstruction and enhancement, focused on preserving anatomical structures and improving image quality.
- **Outcome:** First author of *"ReFineMRI"*, submitted to *IEEE Transactions on Computational Imaging* (2026).

Research Collaboration Apr 2024 – Sep 2024

Universidad de Santiago de Compostela | Santiago de Compostela, Spain
With Dr. Belén Serrano-Antón and Prof. Alberto P. Muñuzuri • Nonlinear Physics Group, CITMAga, FlowReserve Labs S.L.

- Worked on medical imaging research focused on metal artifact reduction and CT domain transformation.
- **Outcome:** First author of *"MAR-DTN"*, published at the 27th International Conference on Pattern Recognition (ICPR 2024).

Publications

Journal Articles

[1] A. Corso, N. Martinel, **M. Rehman**, J. Stancanello, C. Micheloni, C. Deana, C. Cappelletto, P. Chiovati, R. Spizzo, G. Fanetti, A. Dassie, M. Avanzo, "Validation of Synthetic Megavoltage Computed Tomography (MVCT) for Dose Calculation in Radiotherapy Treatment Planning," *Cancers*, vol. 18, no. 10, art. 1603, May 2026. [DOI]

[2] D. Spina, K. Roitero, S. Mizzaro, V. Della Mea, F. Da Ros, M. Soprano, **M. Rehman**, et al., "Report on the Hands-On PhD Course on Responsible AI from the Lens of an Information Access Researcher," *ACM SIGIR Forum*, vol. 58, no. 2, pp. 1–61, Mar. 2025. [PDF]

Journal Abstracts

[3] M. Avanzo, A. Corso, **M. Rehman**, C. Micheloni, N. Martinel, R. Spizzo, C. Deana, G. Fanetti, C. Cappelletto, P. Chiovati, G. Pirrone, S. Marković Perić, A. Dassie, "Evaluation of Quality of Synthetic Megavolt Computed Tomography (MVCT) for Planning of Radiotherapy Treatments of Head and Neck," *Physica Medica: European Journal of Medical Physics*, vol. 142, Feb. 2026. [PDF]

Conference Papers

[4] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "H3D-MarNet: Wavelet-Guided Dual-Path Learning for Metal Artifact Suppression and CT Modality Transformation for Radiotherapy Workflows," in *Pattern Recognition – ICPR 2026*, LNCS vol. 16821, Springer, pp. 123–138, 2026. [arXiv] [Certificate]

[5] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "ReMAR-DS: Recalibrated Feature Learning for Metal Artifact Reduction and CT Domain Transformation," *23rd International Conference on Image Analysis and Processing (ICIAP)*, Sep. 2025. [DOI] [arXiv] [Certificate]

[6] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "MAR-DTN: Metal Artifact Reduction using Domain Transformation Network for Radiotherapy Planning," *27th International Conference on Pattern Recognition (ICPR)*, 2024. [DOI] [arXiv] [Certificate]

[7] **M. Rehman**, C.-M. Own, Z. Li, "A Fast and Robust Embedded Optical Flow Unit on Motion Estimation," in *Proc. 2021 10th International Conference on Software and Computer Applications (ICSCA)*, ACM, pp. 180–185, 2021. [DOI] [Certificate]

Under Review

[8] **M. Rehman**, D. Lyu, M. Staring, N. Martinel, M. Doneva, C. Micheloni, Y. Dong, "ReFineMRI: Structure-Preserving Low-Field MRI Enhancement via Progressive Refinement," submitted to *IEEE Transactions on Computational Imaging (TCI)*, 2026.

[9] **M. Rehman**, N. Martinel, M. Avanzo, R. Spizzo, C. Micheloni, "WAVEFORM-Net: Wavelet-Enhanced Frequency Fusion for Transformer-Based Metal Artifact Reduction in CT Domain Translation," submitted to *Multimedia Tools and Applications* (Springer), 2026.

Poster Presentations

- **ICPR 2026 – H3D-MarNet** – International Conference on Pattern Recognition, Lyon, France [View Poster] Aug 2026
- **MISP^2** – Medical Imaging Symposium for PhDs and Postdocs, LUMC, Leiden, Netherlands [View Poster] Jan 2026
- **ICIAP 2025 – ReMAR-DS** – Int. Conf. on Image Analysis and Processing, Rome, Italy [View Poster] Sep 2025
- **ICVSS 2024** – International Computer Vision Summer School, Sicily, Italy [View Poster] Jul 2024

Technical Skills

- **Programming:** Python, C/C++
- **AI & Computer Vision:** PyTorch, OpenCV, NumPy, Pandas, scikit-image, Matplotlib, Weights & Biases (W&B)
- **Core Expertise:** Deep Learning, Computer Vision, Medical Image Analysis, Generative AI, CNNs, Vision Transformers (ViTs), Diffusion Models, Flow Matching, Image Reconstruction, Image Enhancement, Cross-Modality Domain Translation
- **Medical Imaging:** CT, MRI, DICOM, 3D Slicer (advanced), ITK-SNAP (basic)
- **Tools & Platforms:** Git, PyCharm, Jupyter Notebook / JupyterLab, Draw.io, Jira, Miro
- **Research & Documentation:** LaTeX, BibTeX, EndNote, Mendeley

Professional Experience

Research Collaborator — Sep 2023 – Sep 2026
IRCCS Centro di Riferimento Oncologico di Aviano | Aviano, Italy

- Collaborated with oncologists and medical physicists on CT imaging datasets.
- Developed AI-driven methods for Metal Artifact Reduction (MAR) and cross-modality domain transformation (kVCT $\rightarrow$ MVCT).
- Integrated deep learning models for CT reconstruction and enhancement into radiotherapy workflows.

IT Project Lead and CRM Manager — May 2021 – Jun 2023
Moreno Holdings Pvt Ltd. | Islamabad, Pakistan

Senior Research Executive — Oct 2016 – Dec 2019
Empower Pakistan | Pakistan (Remote, Part-time)

Intern — Oct 2015 – May 2016
Crevo Technologies | Islamabad, Pakistan

Intern — Aug 2015 – Sep 2015
Ministry of Defence Production, Directorate of Procurement (Navy) | Rawalpindi, Pakistan

Awards and Honours

- **PhD Fellowship** – Fully funded doctoral fellowship from the Italian Ministry of Health, in collaboration with the National Cancer Research Institute, CRO IRCCS. 2023
- **Master's Scholarship** – Tianjin University International Student Scholarship, for academic excellence. 2017
- **Top Presenter Award** – Edgestone: A New Era Training Program, METANOIA, Islamabad. 2021
- **Laptop Award** – Prime Minister's Best Student Laptop Scheme, Government of Pakistan. 2015

Conferences, Summer Schools and Research Training

- **ICTCS 2026, 27th Italian Conference on Theoretical Computer Science** – Udine, Italy Sep 2026
- **RISE-MICCAI Summer School 2025** – MICCAI Society Jul 2025
- **International Summer School on AI (AI-DLDA), 8th Edition** – University of Udine Jun 2025
- **International Computer Vision Summer School (ICVSS 2024)** – Sicily, Italy [Certificate] Jul 2024
- **International Summer School on AI (AI-DLDA), 7th Edition** – University of Udine Jul 2024
- **Summer School for Humans and Technology: Technologies for Good** – DAAD, Germany [Certificate] Sep 2023

Professional Service and Affiliations

- **Peer Reviewer** – *Pattern Recognition* (Elsevier); International Conference on Image Analysis and Processing (ICIAP).
- **Member**, Italian Association for Research in Computer Vision, Pattern Recognition and Machine Learning (CVPL), affiliated with the IAPR. 2025 – Present

Languages

- **English** – Fluent; IELTS Band 7 (2017, 2023); Ecctis UK recognised (2026)
- **Urdu** – Native
- **Chinese** – Basic (HSK 1)

Certifications

Specializations

- **Edgestone: A New Era Training Program** – METANOIA, Islamabad [Verify] — Apr – May 2021
  - Top Presenter Award [Verify] — 2021
- **Major Engineering Project Performance** – University of Leeds (ID ZKT8UBFZNHHF) [Verify] — Jul 2020
- **Engineering Project Management Specialization** – Rice University (ID 5ZAL4FGYD9HJ) [Verify] — Jun 2020
  - Risk, Quality, Teams, and Procurement [Verify] — Jun 2020
  - Scope, Time and Cost Management [Verify] — Jun 2020
  - Initiating and Planning [Verify] — Mar 2020

Transversal Courses

- **Introduction to Intellectual Property (IP) and Patents** (ID 3404D57-1275FC554-31210A1) [Verify] — Jul 2026
- **Introduction to Generative AI for Educators** (ID 3404D57-35F1CD5-31210A1) [Verify] — Jul 2026
- **Cyber Security Fundamentals and Best Practices** (ID 3404D57-1275F28A9-31210A1) [Verify] — Jul 2026
- **Writing Research Papers: Structure and Style** (ID 3404D57-34F688B-31210A1) [Verify] — Jul 2026
- **Introduction to Agentic AI: Principles and Applications** (ID 3404D57-1275F798C-31210A1) [Verify] — Jul 2026
- **Streamline Your Research Writing with AI** (ID 3404D57-34F65C9-31210A1) [Verify] — Jul 2026

Courses and Certificates

- **WordPress Course, Batch 8** – DigiSkills.pk (ID US7KVWGPQ) [Verify] — Jan 2021
- **Freelancing Course, Batch 8** – DigiSkills.pk (ID 2YCXAEEPQ) [Verify] — Jan 2021
- **Professional Scrum Master** – Pakistan Software Export Board [Verify] — Jul 2020
- **Email Marketing Certified** – HubSpot Academy (ID 0797555fb3a34798a7720ae79118bc52) — Jul 2020
- **Ufone Master Class: Career Path Management** – Ufone 5G — Jul 2020
- **Ufone Master Class: MS PowerPoint Essentials** – Ufone 5G — Jul 2020
- **Ufone Master Class: Microsoft Excel Essentials** – Ufone 5G — Jul 2020
- **Scrum: The Basics** – LinkedIn Learning [Verify] — Jun 2020
- **Photoshop 2020 Essential Training** – LinkedIn Learning (ID AbX66qxUhMqA2Dez0F6wBux_rSk5) [Verify] — Jun 2020
- **Positive Psychology** – University of North Carolina at Chapel Hill (ID G34KFTRQKXT2) [Verify] — Apr 2020
- **Chinese for Beginners** – Peking University (ID CYWDDN7A56JH) [Verify] — Apr 2020
- **Project Management: The Basics for Success** – University of California, Irvine (ID VPFKJFUELMA7) [Verify] — Mar 2020
- **Women in Leadership: Inspiring Positive Change** – Case Western Reserve University [Verify] — Mar 2020
- **Successful Negotiation: Essential Strategies and Skills** – University of Michigan [Verify] — Mar 2020
- **Managing Project Risks and Changes** – University of California, Irvine (ID 6D3464E3295V) [Verify] — Jan 2020
- **Budgeting and Scheduling Projects** – University of California, Irvine (ID UMFW83286QQC) [Verify] — Jan 2020
- **Business Management Technology: IT Project Management** – Indian School of Business [Verify] — Dec 2019
- **Initiating and Planning Projects** – University of California, Irvine (ID EV9AQK4M4E5M) [Verify] — Dec 2019
- **Fundamentals of Project Planning and Management** – University of Virginia (ID LSTTEQVTA4LF) [Verify] — Dec 2019
- **IEEE Pakistan Student / Young Professionals / Women in Engineering Congress, 9th Episode** – COMSATS University Islamabad — Jul 2015
- **CPD Workshop: Future of Parallel Processing in IT Industry** – Fatima Jinnah Women University — Jan 2015
- **CPD Workshop on XML Learning** – Fatima Jinnah Women University — Jan 2014

Volunteering and Activities

- **Soprano Singer** (Nov 2019) – International Students Chorus, Tianjin University; performed to represent cultural exchange through music.
- **Cultural Volunteer** (Oct 2018) – Chinese Opera Culture Week (Peking Opera), 9th China (Beijing) International Garden Expo.
- **Volunteer** – Esmile Khursheed Welfare, Rawalpindi; ran extra-curricular and confidence-building sessions with students.
- **Volunteer** – Al-Shifa Eye Trust Hospital, Rawalpindi; assisted staff with patient handling during daily operations.

Hobbies & Interests

- **Travel & Exploration:** Discovering new places, experiencing different cultures, collecting souvenirs and exploring local cuisines.
- **Fitness & Outdoors:** Fitness, athletics, gardening and fishing.

References

Available upon request.